

SOW (Statement of Work) – March2026

Introduction – Module 1

Module: Subscription & Tenant Administration

Product: Suri LMS Cloud

Version: 1.0

Document Type: Engineering SOW

Target Audience: Product, Engineering, QA, DevOps

Objective

The Subscriber Module enables:

- Learners, Teachers, and Organizations to subscribe to LMS plans
- Secure online payments (monthly/annual)
- Discount and coupon handling
- Tenant-level admin panel
- Role-based user management
- Access control to LMS features
- Subscription lifecycle management
- Integration with Core LMS Engine

This module must support **multi-tenant SaaS architecture**.

- Enable recurring SaaS revenue model
- Support individual and organization subscriptions
- Provide full tenant-level administration
- Enforce feature-level access control
- Allow plan-based LMS feature restrictions
- Ensure secure, scalable billing operations

Users and Scope

User	Description
Individual Learner	Personal subscription
Teacher	Course creator subscription
Organization Admin	Company-level subscription (Enterprise)
Organization User	Managed under org subscription (Enterprise)
Super Admin	Platform owner (Cloud Admin)

The Subscriber Module includes:

1. Subscription Plan Management
2. Pricing & Discount Engine
3. Payment Processing
4. Subscription Lifecycle Management
5. Tenant Admin Panel
6. Role & User Management
7. Feature Access Control
8. LMS Module Integration
9. Billing & Invoice Management
10. Reporting & Analytics

Requirements-Functional

EPIC 1: Subscription Plan Management

Objective:

Allow Super Admin to define and manage subscription plans.

Features:

- Create plan (Basic, Pro, Enterprise)
- Monthly / Annual billing cycle
- Feature mapping per plan
- User limit per plan
- Storage limit
- Course creation limit
- API usage limit (AI usage cap)
- Enable/disable plan

- Plan versioning

User Stories:

- As Super Admin, I can create a subscription plan.
- As Super Admin, I can define feature access per plan.
- As Super Admin, I can configure annual discount percentage.
- As Super Admin, I can deactivate a plan.

Acceptance Criteria:

- Plan must not be deletable if active subscribers exist.
- Changes must not impact existing subscribers immediately (version control required).

EPIC 2: Subscription Purchase & Payment**Objective:**

Allow users to subscribe and make payments securely.

Features:

- Plan selection UI
- Coupon code validation
- Tax calculation
- Multi-currency support
- Payment gateway integration
- Invoice generation
- Auto-renewal
- Upgrade/downgrade
- Pro-rata calculation

User Stories:

- As a learner, I can choose a plan.
- As a teacher, I can subscribe annually and get discount.
- As an organization, I can purchase enterprise plan.
- As user, I receive invoice after payment.
- As user, I can cancel auto-renewal.

Acceptance Criteria:

- Payment must be verified server-side.

- Webhook validation required.
- No access until payment confirmation.

EPIC 3: Subscription Lifecycle Management

States:

Trial -> Active -> Grace Period -> Expired -> Cancelled -> Suspended

Functional Requirements:

- Trial period management
- Renewal reminders (email + notification)
- Grace period after expiry
- Auto-suspension if unpaid
- Manual override by Super Admin

Acceptance Criteria:

- System must revoke feature access upon expiry.
- Grace period configurable globally.

EPIC 4: Tenant Admin Panel

Objective:

Provide organization-level management console.

Features:

- Dashboard (usage stats)
- User management
- Role management
- Department management
- Course assignment
- Subscription status overview
- Payment history
- Invoice download

User Stories:

- As Org Admin, I can create users.
- As Org Admin, I can assign roles.
- As Org Admin, I can deactivate users.

- As Org Admin, I can see subscription usage.

EPIC 5: Role & Access Management (RBAC)

Roles:

- Org Admin
- Teacher
- Learner
- Moderator
- Custom Role

Features:

- Create custom role
- Assign permissions
- Feature-level control
- API access control
- Role audit log

Access Mapping:

Each role must map to:

- Course creation
- Quiz management
- Forum moderation
- Badge issuance
- Report access
- AI usage access

Acceptance Criteria:

- Role changes must reflect immediately.
- Unauthorized access must return 403.

EPIC 6: LMS Core Feature Integration

The subscription module must integrate with:

LMS Feature	Subscription Check Required
Course Creation	Yes

LMS Feature	Subscription Check Required
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Quiz Creation	Yes
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Certification	Yes
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Badge Issuance	Yes
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Forum Posting	Yes
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AI Tutor Usage	Yes
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Analytics Dashboard	Yes
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System Behavior:

On every API call, validate:

- Subscription active?
- Plan allows this feature?
- User role authorized?

EPIC 7: Billing & Invoicing

Features:

- Invoice PDF generation
- GST/VAT support
- Payment history
- Refund processing
- Credit note management

Compliance:

- Invoice number auto-sequencing
- Timezone safe timestamping

EPIC 8: Reporting & Monitoring

Reports:

- Active subscribers
- Revenue per plan
- Churn rate
- Upgrade/downgrade stats

- Organization usage

Requirements-Technical

Architecture

React Frontend



Node API Gateway



Subscription Microservice



PostgreSQL/MySQL



Payment Gateway API

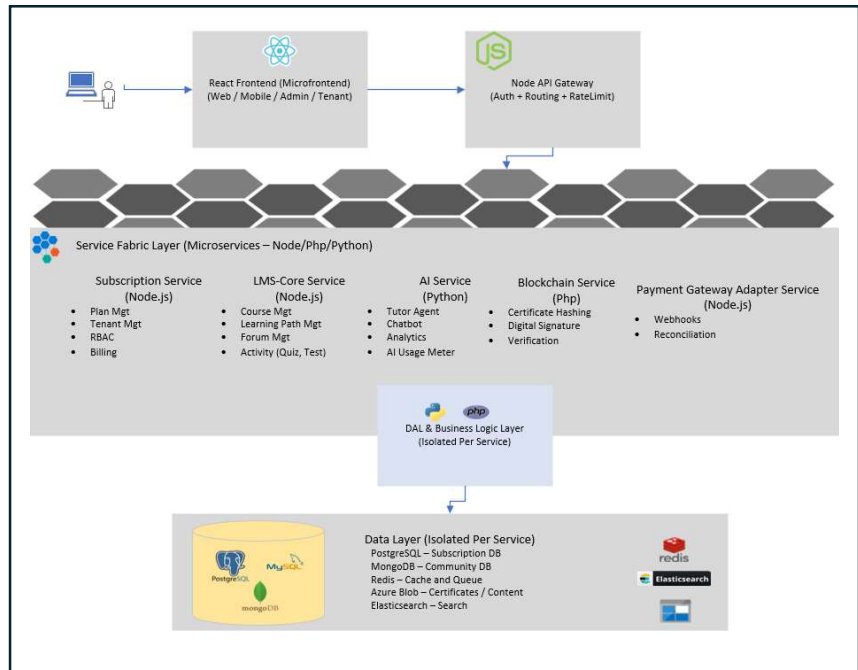
Database Tables (High-Level)

subscription_plans

- id
- name
- price_monthly
- price_annual
- features_json
- max_users
- max_courses
- ai_limit

subscriptions

- id
- tenant_id
- plan_id
- status
- start_date
- end_date
- auto_renew
- billing_cycle



subscription_payments

- id
- subscription_id
- transaction_id
- amount
- currency
- status

tenant_users

- id
- tenant_id
- role_id
- status

Requirements- Non-Functional

Security

- JWT authentication
- Role-based authorization
- Encrypted payment data
- Webhook signature validation
- Audit logging

Performance

- Subscription check <100ms
- Payment confirmation <3 seconds

Scalability

- Support 100k+ subscribers
- Horizontal scaling

Requirements- Integration

Must expose API for:

- Subscription status check
- Feature entitlement check
- Role validation
- Plan details retrieval

Must integrate with:

- Core LMS module
- AI usage meter
- Blockchain certificate issuance (only active subscription allowed)

Deliverables & Acceptance Criteria

- Fully functional subscription engine
- Tenant admin dashboard
- Payment integration
- Role management system
- API documentation
- Database schema
- Deployment scripts
- Unit & integration test coverage

Acceptance Criteria:

- Subscription purchase success rate > 99%
- Access revocation accuracy 100%
- Payment reconciliation accuracy 100%
- Multi-tenant data isolation validated

Out of Scope:

- Multi-currency wallet
- Multi-country taxation engine
- Custom enterprise billing contracts